Marking Scheme

2026 · Paper 1 · 105 marks

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating： **3.5** / 5.0

General Rules
• M / A / PROOF mark types
• **f.t.**：承上題錯誤但方法正確者，可獲後續步驟分。
• Deduction cap： A(1)=1 · A(2)=1 · B=0 (u, pp)
• u = 單位遺漏或錯誤；pp = 準確度不符要求。每部分最多扣 1 分。
• 答案正確且步驟合理者可獲滿分。
• 符號書寫略有出入但意思清楚者不扣分。 · Benefit of doubt
• 分數、小數或根式形式均接受，除非題目另有指明。

Section A(1) (35 marks)

1. Algebraic Fractions

a (3 marks)

M1 $= \dfrac{4(x+1)-(3x-2)}{(3x-2)(x+1)}$

M1 $= \dfrac{4x+4-3x+2}{(3x-2)(x+1)}$

A1 $= \dfrac{x+6}{(3x-2)(x+1)}$

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2. Change of Subject

a (3 marks)

M1 $2(y+b)=5(y-a)$

M1 $2y+2b=5y-5a \Rightarrow 3y=5a+2b$

A1 $y=\dfrac{5a+2b}{3}$

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3. Factorization

a (1 mark)

A1 $(2p-3q)(p+2q)$

b (2 marks)

M1 $=(2p-3q)(p+2q)-3(2p-3q)$ f.t.

A1 $=(2p-3q)(p+2q-3)$

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4. Inequalities

a (3 marks)

M1 $3x+1>5x-15 \Rightarrow x<8$

M1 $4-x\le 9 \Rightarrow x\ge -5$

A1 $-5\le x<8$

b (1 mark)

A1 -5 f.t.

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5. Rates, Ratios and Proportions

a (4 marks)

M1 $Let a = 2k and b = 5k$

M1 $2k + c = 3(5k - c) \Rightarrow 4c = 13k$

A1 $c = \frac{13k}{4}$

A1 $\frac{a+b}{c} = \frac{7k}{\frac{13k}{4}} = \frac{28}{13}$

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6. Percentages

a (4 marks)

M1 $Let \x be the cost; marked price = \1.4x$

M1 $1.4x(1 - 15\%) - x = 57$

A1 $0.19x = 57, x = 300$

A1 $Marked price = \420$

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7. Polar Coordinates

a (1 mark)

A1 $\angle AOB = 110° - 20° = 90°$

b (1 mark)

A1 $\angle AOC = 200° - 20° = 180°, so A, O and C are collinear.$

c (2 marks)

M1 $AC = 4 + 6 = 10; \angle BOC = 90°, AB = \sqrt{4^2 + 3^2} = 5, BC = \sqrt{3^2 + 6^2} = 3\sqrt{5}$ f.t.

A1 $Perimeter = 15 + 3\sqrt{5}$ f.t.

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8. Congruent Triangles

a (2 marks)

M1 $\angle ABC = \angle AED = 90°, AC = AD (hypotenuses) and BC = ED (given)$

A1 $\therefore \triangle ABC \cong \triangle AED (RHS)$

b (3 marks)

M1 $AC = \sqrt{8^2 + 6^2} = 10cm = AD = CD, so \triangle ACD is equilateral$

M1 $Area = 2 \times \frac{1}{2}(8)(6) + \frac{1}{2}(10)(10)\sin 60°$ f.t.

A1 $= (48 + 25\sqrt{3})cm^2$ f.t.

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9. Measures of Central Tendency and Probability

a (2 marks)

M1 $\frac{k}{15 + k} = \frac{2}{5}$

A1 $k = 10$

b (3 marks)

A1 $Mean = \frac{46}{25} = 1.84$ f.t.

A1 $Mode = 2$ f.t.

A1 $Median = 2$ f.t.

Section A(2) (35 marks)

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10. Variations

a (3 marks)

M1 $g(x) = a + bx, where a, bare non-zero constants$

M1 $a + 2b = -1 and a + 5b = 8$

A1 $g(x) = 3x - 7$

b (3 marks)

M1 $3x^2 - 7x + k = 0$ f.t.

M1 $\Delta = 49 - 12k \geq 0$ f.t.

A1 $k \leq \frac{49}{12}$ f.t.

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11. Measures of Dispersion

a (3 marks)

M1 $Sum = 276 + m + n$

M1 $\frac{276 + m + n}{11} = 26$

A1 $n = 10 - m$

b (1 mark)

A1 $2 \leq m \leq 5 (since n = 10 - m \leq 8); least range when m = 5 (n = 5) : 39 - 15 = 24 hours$ f.t.

c (3 marks)

M1 $Q_1 = 18, Q_3 = 30 + n, inter-quartile range = 12 + n$ f.t.

M1 $1 \leq n \leq 8 and n = 10 - m with 0 \leq m \leq 5, so 5 \leq n \leq 8$ f.t.

A1 $Greatest inter-quartile range = 12 + 8 = 20 hours$ f.t.

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12. Coordinate Geometry

a (3 marks)

M1 $Let the side be t; S = \left(\frac{t}{m}, t\right)$

M1 $R = \left(\frac{t}{m} + t, t\right)$

A1 $Slope of OR = \frac{t}{\frac{t}{m} + t} = \frac{m}{1 + m}$

b (4 marks)

M1 $OA : y = 2x, so by (a), R lies on y = \frac{2}{3}x$ f.t.

A1 $AB : y = -x + 9$

M1 $\frac{2}{3}x = -x + 9$ f.t.

A1 $R = \left(\frac{27}{5}, \frac{18}{5}\right)$ f.t.

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13. Similar Solids

a (3 marks)

M1 $Volume of original pyramid = \frac{1}{3}(12^2)(8) = 384cm^3$

M1 $Small pyramid: base side 6cm, height 4cm, volume = 48cm^3$

A1 $Volume of X = 336cm^3$

b (4 marks)

M1 $Slant height of each lateral face of X = \sqrt{4^2+3^2} = 5cm$

A1 $Total surface area of X = 12^2 + 6^2 + 4 \times \frac{1}{2}(12+6)(5) = 360cm^2$ f.t.

M1 $If similar, ratio of areas = \left(\frac{6}{4}\right)^2 = \frac{9}{4}, so the total surface area of Y would be 810cm^2$ f.t.

A1 $810 \neq 800, so X and Y are not similar.$ f.t.

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14. Polynomials

a (1 mark)

A1 $-3b = -6 \Rightarrow b = 2$

b(i) (4 marks)

M1 $F(-3) = (11-3a)(6-3c) = 0$ f.t.

A1 $Since a is an integer, 11-3a \neq 0, so c = 2$

M1 $F(2) = (6+2a)(4+4-3) = -10$ f.t.

A1 $a = -4$

b(ii) (3 marks)

M1 $F(x) = (x^2-4x+2)(x+3)(x-1)$ f.t.

M1 $x^2-4x+2 = 0 \Rightarrow x = 2 \pm \sqrt{2}$ f.t.

A1 $The roots are -3, 1, 2+\sqrt{2}, 2-\sqrt{2}; there are 2 irrational roots.$ f.t.

Section B (35 marks)

15. Logarithmic Functions

a (3 marks)

M1 $\log_2 y - 5 = -3(\log_2 x - 1) \Rightarrow \log_2 y = 8 - 3\log_2 x$

M1 $\log_2 y = \log_2 \frac{2^8}{x^3}$

A1 $y = \frac{256}{x^3}$

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16. Probability

a (2 marks)

M1 $\frac{C_1^5 C_2^4}{C_3^9}$

A1 $= \frac{30}{84} = \frac{5}{14}$

b (2 marks)

M1 $\frac{C_3^4}{C_3^9} + \frac{5}{14}$ f.t.

A1 $= \frac{17}{42}$ f.t.

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17. Locus and Equations of Circles

a(i) (1 mark)

A1 $\Gamma is the perpendicular bisector of AB$

a(ii) (2 marks)

M1 $Mid-point of AB = (4, 5); slope of AB = -1, so slope of \Gamma = 1$ f.t.

A1 $x - y + 1 = 0$

b(i) (3 marks)

M1 $The centre G lies on \Gamma and on the perpendicular bisector of BC, x = 2$ f.t.

A1 $G = (2, 3), radius^2 = (1-2)^2 + (8-3)^2 = 26$ f.t.

A1 $x^2 + y^2 - 4x - 6y - 13 = 0$ f.t.

b(ii) (2 marks)

M1 $\angle TUG = \angle TVG = 90°, so TG is a diameter of the circumcircle; TG = 10, radius = 5$ f.t.

A1 $Area = 25\pi \approx 78.5 < 80; no.$ f.t.

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18. Trigonometry in 3-D

a(i) (2 marks)

M1 $AC^2 = 10^2 + 12^2 - 2(10)(12)\cos 70°$

A1 $AC \approx 12.7cm$

a(ii) (2 marks)

M1 $\dfrac{\sin \angle BAC}{12} = \dfrac{\sin 70°}{AC}$ f.t.

A1 $\angle BAC \approx 62.4°$

b(i) (2 marks)

M1 $Distance from B to AC = AB \sin \angle BAC \approx 8.862cm; shortest distance = 8.862 \sin 60°$ f.t.

A1 $\approx 7.67cm$ f.t.

b(ii) (2 marks)

M1 $The shortest distance between B and the plane ACD is about 7.67cm$ f.t.

A1 $7.67 < 8, so there are points on the plane less than 8cm from B; I disagree.$ f.t.

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19. Quadratic Functions and Sequences

a (2 marks)

M1 $f(x) = (x - 2a)^2 + b$

A1 $V_1 = (2a, b)$

b (2 marks)

A1 $Enlarge the graph horizontally to 2 times the original (along the x-axis), then translate upwards by 7 units$

A1 $V_2 = (4a, b + 7)$ f.t.

c(i) (4 marks)

M1 $2b = 2a + 4a \Rightarrow b = 3a$

M1 $(4a)^2 = b(b + 7) = 3a(3a + 7)$ f.t.

M1 $7a^2 - 21a = 0$ f.t.

A1 $a = 3 (since a > 0), b = 9$ f.t.

c(ii) (4 marks)

M1 $V_1 = (6, 9), V_2 = (12, 16); \overrightarrow{V_1V_2} = \overrightarrow{RS} = (6, 7), so V_1V_2SR is a parallelogram (if not degenerate) with V_1V_2 = \sqrt{85}$ f.t.

M1 $For a rhombus, RV_1 = \sqrt{85} : (t - 6)^2 + (t + 11)^2 = 85$ f.t.

M1 $2t^2 + 10t + 72 = 0, \Delta = 100 - 576 < 0$ f.t.

A1 $No real t exists, so it is not possible.$ f.t.

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